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ORIGINAL ARTICLE

Climatic limits for the present distribution of beech (*Fagus* L.) species in the world

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ABSTRACT

Aim Beech (*Fagus* L., Fagaceae) species are representative trees of temperate deciduous broadleaf forests in the Northern Hemisphere. We focus on the distributional limits of beech species, in particular on identifying climatic factors associated with their present range limits.

Location Beech species occur in East Asia, Europe and West Asia, and North America. We collated information on both the southern and northern range limits and the lower and upper elevational limits for beech species in each region.

Methods In total, 292 lower/southern limit and 310 upper/northern limit sites with available climatic data for all 11 extant beech species were collected by reviewing the literature, and 13 climatic variables were estimated for each site from climate normals at nearby stations. We used principal components analysis (PCA) to detect climatic variables most strongly associated with the distribution of beech species and to compare the climatic spaces for the different beech species.

Results Statistics for thermal and moisture climatic conditions at the lower/southern and upper/northern limits of all world beech species are presented. The first two PCA components accounted for 70% and 68% of the overall variance in lower/southern and upper/northern range limits, respectively. The first PCA axis represented a thermal gradient, and the second a moisture gradient associated with the world-wide distribution pattern of beech species. Among thermal variables, growing season warmth was most important for beech distribution, but winter low temperature (coldness and mean temperature for the coldest month) and climatic continentality were also coupled with beech occurrence. The moisture gradient, indicated by precipitation and moisture indices, showed regional differences. American beech had the widest thermal range, Japanese beeches the most narrow; European beeches occurred in the driest climate, Japanese beeches the most humid. Climatic spaces for Chinese beech species were between those of American and European species.

Main conclusions The distributional limits of beech species were primarily associated with thermal factors, but moisture regime also played a role. There were some regional differences in the climatic correlates of distribution. The growing season temperature regime was most important in explaining distribution of Chinese beeches, whilst their northward distribution was mainly limited by shortage of precipitation. In Japan, distribution limits of beech species were correlated with summer temperature, but the local dominance of beech was likely to be dependent on snowfall and winter low temperature. High summer temperature was probably a limiting factor for southward extension of American beech, while growing season warmth seemed critical for its northward distribution. Although the present distribution of beech species corresponded

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well to the contemporary climate in most areas, climatic factors could not account for some distributions, e. g., that of *F. mexicana* compared to its close relative *F. grandifolia*. It is likely that historical factors play a secondary role in determining the present distribution of beech species. The lack of *F. grandifolia* on the island of Newfoundland, Canada, may be due to inadequate growing season warmth. Similarly, the northerly distribution of beech in Britain has not reached its potential limit, perhaps due to insufficient time since deglaciation to expand its range.

Keywords

Climatic index, climatic space, continentality, *Fagus*, growing season warmth, precipitation, principal components analysis, range limit, temperate forest.

INTRODUCTION

Beech (*Fagus* L., Fagaceae) are among the most representative trees in the temperate deciduous broadleaf forests of the Northern Hemisphere (Shen, 1992; Denk, 2003). The genus includes ten primary species and two minor segregates (Willis, 1966) broadly distributed in three isolated regions: East Asia, Europe and West Asia, and North America. Six species occur in East Asia (*F. engleriana*, *F. longipetiolata*, *F. lucida* and *F. hayatae* in China and *F. crenata* and *F. japonica* in Japan) (Horikawa, 1972; Editorial Committee for Flora of China, 1999), two in Europe and West Asia (*F. sylvatica* and *F. orientalis*) (Jalas & Suominen, 1972–91), and one in North America (*F. grandifolia*) (Little, 1965, 1979). Some studies have recognized two additional beech species, one on a small island off Korea: *F. multinervis* (Kim *et al.*, 1986; Kim, 1988) and another one in the north-eastern mountains of Mexico: *F. mexicana* (e.g. Miranda & Sharp, 1950; Rzedowski, 1983; Maycock, 1994; Peters, 1995). *Fagus multinervis* is a segregate of *F. engleriana* in China (Okubo *et al.*, 1988; Peters, 1992; Shen, 1992; Denk, 2003); *F. mexicana* is a segregate of the North American *F. grandifolia*. For the purposes of the present study we accept the validity of both segregate species.

In East Asia, beech occurs primarily in mountain areas. All Chinese beeches are restricted to remote subtropical/warm-temperate mountain areas, ranging from south China to the Yangtze River (*c.* 33° N), and from the south-east coast of the East China Sea to the eastern edge of the Tibetan Plateau (Tsien *et al.*, 1975; Wu, 1980; Hou, 1988; Hong & An, 1993; Cao *et al.*, 1995). Compared with the continuous distribution of other Chinese beeches, *F. hayatae* is isolated in three very limited mountain areas: north-east Taiwan, eastern Zhejiang and north-west Sichuan. Unlike beech in other regions, Chinese beeches rarely form pure stands but typically occur mixed with various deciduous broad-leaved trees (*Betula*, *Acer*, *Liriodendron*, *Davidia*, *Tilia*, *Carpinus* and *Nyssa*), evergreen broad-leaved trees (*Lithocarpus*, *Cyclobalanopsis*, *Manglietia* and *Castanopsis*) and evergreen needle-leaved (*Tsuga*) trees (Wu, 1980; Hou, 1983, 1988; Hsieh, 1989; Cao, 1995; Fang *et al.*, 1996).

Two beech species, *F. crenata* and *F. japonica*, are native to Japan. The former is distributed from Kyushu (*c.* 30.5° N) to southern Hokkaido (*c.* 42.8° N), whereas the latter is limited to the south of Iwate-ken (Horikawa, 1972; Miyawaki, 1980–89). Beech forest in Japan falls into two broad forest types: the Pacific-Ocean-side type that grows intermixed with many other temperate tree species, and the Japan-Sea-side type where beech is frequently dominant (Yamazaki, 1983; Maeda, 1991). Forests with *F. multinervis* are restricted to a small island, Ulreung-do in South Korea, and they are more or less similar in community composition and structure to Japanese beech forests of the Japan-Sea-side type (Kim *et al.*, 1986; Kim, 1988).

In North America, *F. grandifolia* is one of the most widespread species among temperate trees, covering almost all the temperate zone along the Appalachian Mountains from the northern edge of the subtropical zone almost to the southern edge of the boreal zone (USDA Forest Service, 1975). A number of studies have shown large differences in community composition and structure in different climatic regions (Braun, 1967; Barnes, 1991; Maycock, 1994). Based on differences in geographical distribution and morphological characters, three races of *F. grandifolia* (grey beech, white beech, and red beech) have been identified (Camp, 1951; Braun, 1967; Maycock, 1994) but these have never been given species status.

Another beech species in North America, *F. mexicana*, is found in only four montane localities in north-eastern Mexico (Little, 1965; Rzedowski, 1983). Its community characters are more or less similar to those of Chinese beech forests, usually mixing with many species of *Quercus*, *Magnolia*, *Acer* and *Carya* (Miranda & Sharp, 1950; Rzedowski, 1983; Peters, 1995; Williams-Linera *et al.*, 2000).

In Europe, *F. sylvatica* spreads from Sicily in southern Italy (*c.* 37.7° N) to Bergen in south Norway (*c.* 60.7° N) (Jalas & Suominen, 1972–91; Feoli & Lagonegro, 1982; Jahn, 1991); this is the most widely distributed of all beech species. *Fagus orientalis* replaces *F. sylvatica* in a small region of southeast Europe and spreads into West Asia: northern Turkey, the Caucasus and the Elburz Mountains of northern Iran, where the climate is more or less continental (Jalas & Suominen, 1972–91; Davis, 1982).

The three regions where beech species occur are controlled by different air masses and have various topographic settings. A monsoon climate prevails in East Asia with cold and dry air masses from Siberia predominant in winter and hot, humid subtropical highs from the Pacific Ocean predominant in summer. This causes abundant summer rainfall and high temperature after a dry and cold winter (Arakawa, 1969; Shen, 1986). The Himalayan Range, higher to the west and lower eastwards in mainland China, intensifies this climatic difference between winter and summer (Chang, 1983; Fang *et al.*, 1996) on the continent in comparison to the Japanese archipelago.

In eastern North America, the climate is governed by two major circulations: cold and dry air masses from the Arctic and moist warm air masses from the southern tropical seas, and therefore the climatic patterns resemble more or less those in East Asia. However, there is no topographic barrier akin to the Himalayas to block meridional air mass exchange (Lydolph, 1985).

The European and West Asian region has a narrower seasonal cycle of temperatures and rainfall than the two other regions. In Europe, the air masses resemble those of western Northern America, but are much influenced by topography. The east–west trend of European mountain ranges reduces northward invasion of large subtropical bodies of warm air, and the Mediterranean Sea also plays a role in the generation and routes of cyclonic storms (Lydolph, 1985).

The floristic, climatic and topographic patterns of these regions where beech occurs have attracted the attention of previous investigators. For example, using *Fagus* pollen data and monthly mean temperature, Huntley *et al.* (1989) studied climatic control of beech distribution and abundance in Europe and North America. Peters (1992) and Peters & Poulson (1994) compared tree growth, and community structure and dynamics of the world beech species. Maycock (1994) documented detailed information on differences in community composition between North American and Japanese beeches. Iverson & Prasad (1998) used forest inventory data to estimate the climate envelope for *F. grandifolia* and predict its range extension under climate change. Piovesan & Adams (2001) compared masting behaviours of beech from Europe, eastern North America and Japan, and discussed their links to climatic variations. In addition, some case studies on relationships between beech distribution and climates have been conducted, but most are restricted to single species in a region (e.g. Birks, 1989; Cao *et al.*, 1995; Sykes *et al.*, 1996).

Taken together, the studies mentioned above provide useful ecological comparisons among beech species in the three regions where they occur, but many questions remain about relationships between beech distribution and climate. For example, how different are the geographical patterns shown in the three separate regions where beech species occur? What climatic factors control such patterns? Can contemporary climate explain the present distribution of beech species? Answering these questions is not only important in biogeography, but also will provide a firmer basis for predicting the effects of climate change on the future distribution of beech species. Focusing on distribution limits (lower or southern and upper or northern limits) of the world beech species, we therefore compare their climatic spaces to detect possible factors limiting their distributions.

DATA AND METHODS

Data sets for beech distribution

Data sets for beech distribution were assembled by reviewing ecological, botanical and geographical journals, books and reports to record the geographical location, lower and upper elevational limits, and associated information (e.g. topography, climate and community characteristics) for each site where a beech species was reported to occur. Vertical distribution ranges were reported especially frequently in East Asia, but less often in Europe and North America where topographic relief at the range limits of beech species is less extreme. We assembled data for *F. grandifolia* from a silvicultural atlas (USDA Forest Service, 1975) and ancillary literature reports. We found relatively little data for *F. sylvatica* and *F. orientalis*; the map of Jalas & Suominen (1972–91) yielded data on northern range limits for these species, but high relief and uncertain elevational distributions left the southern limits undefined.

When latitude and longitude were not reported for study locations, we used gazetteers to georeference the sites: (1) *China Places Name* (Anon., 1986) and *Atlas of Land Use in China* (Editorial Committee of 1/1,000,000 Land-use Map of China, 1990) for Chinese beeches; (2) *Gazetteer to AMS 1 : 250,000 Maps of Japan* (Corps of Engineers, US Army, 1956) and *The National Atlas of Japan* (Geographical Survey Institute, 1977) for Japanese beeches; (3) *American Places Dictionary* (Abate, 1994) for American beech; and (4) *Atlas of the World* (Times, 1992) for others. In total we identified 350 sites with reliable data for the lower/southern limits and 353 sites for the upper/northern limits of beech species (Table 1; Fang, 2003). For detailed information on the location and elevation of distribution limits for each beech species, see Appendix S1 in Supplementary Material.

Climatic data

We used monthly mean temperatures and monthly precipitation records from the following data sources: (1) China Meteorological Agency (1984); (2) Japan Meteorological Agency (1972); (3) Central Meteorological Office of Korea (1972); (4) National Climatic Center, NOAA (1983) for USA; (5) Atmospheric Environmental Service, Environment Canada (1982) for Canada; and (6) Wernstedt (1972) for Mexico, Europe and West Asia. The period of record for most stations was 1951–80 or 1941–71.

Temperatures at altitudes of the upper and lower elevation limits of beech were estimated for each locality

Table 1 Sample size for distribution limits of all world beech species. Data set for *Fagus grandifolia* was mainly extracted from its southern and northern distributional edges, and the northern limit of *F. sylvatica* actually lies to the north-east according to the map of Jalas & Suominen (1972–91)

Species	Distribution	Lower limit	Upper limit
F. crenata	Pacific Ocean side, Japan	42	46
F. japonica	Japanese Sea side, Japan	28	22
F. engleriana	South and southwest China	34	36
F. hayatae	Taiwan, Zhejiang, and Sichuan, China	8	10
F. longipetiolata	South, east, and southwest China	55	55
F. lucida	East, and Central China	60	54
F. multinervis	Ulreung-do, South Korea	1	1
*F. grandifolia**	Eastern North America	80	80
F. mexicana	Northeastern Mexico	4	4
F. sylvatica†	Europe	32	40
F. orientalis	Southeast Europe and West Asia	6	5
Total		350	353

*Most data for southern or northern extremes.
†Most data for northern edge.

using a mean lapse rate of 0.6 °C per 100 m (Barry, 1992) together with data from the nearest climatic station. The rainfall data for each beech site were estimated from relationships between precipitation and elevation regressed by using rainfall and altitude data at more than five climatic stations close to the beech site. Similar to Huntley *et al.* (1989), distances between beech sites and climatic stations were less than 0.5° of latitude and 1.0° of longitude (*c.* 50–100 km radius). In total, we secured reliable climate data for 292 sites at the lower/southern limit and 310 sites at the upper/northern limit of beech species.

Climatic parameters

Growing season warmth

The distribution limits of many tree species are closely related to growing season temperature (e.g. Kira, 1945, 1991; Holdridge, 1947; Tuhkanen, 1980; Woodward, 1987; Prentice *et al.*, 1992; Sykes *et al.*, 1996). We used Kira's Warmth Index (WI) (Kira, 1945, 1991) and Holdridge's annual biotemperature (ABT) (Holdridge, 1947) as proxies for growing season warmth, given respectively by:

$$\mathrm{WI} = \sum (T - 5) \text{ (for months in which } T > 5\,^{\circ}\mathrm{C}) \qquad (1)$$

$$\mathrm{ABT} = \frac{\sum T}{12} \text{ (for months in which } 0 < T < 30\,^{\circ}\mathrm{C}) \qquad (2)$$

Coldness and winter temperature summation

A number of studies have shown the importance of minimum winter temperatures in controlling distributional limits of plant species (e.g. Sakai & Weiser, 1973; Woodward, 1987; Sykes *et al.*, 1996; Pederson *et al.*, 2004). Mean temperature for the coldest month (MTCM) is often used as a surrogate for the minimum winter temperature (Solomon, 1986; Ohsawa, 1990; Prentice *et al.*, 1992; Sykes *et al.*, 1996). In this study, the coefficient of determination (R^2) between these two variables was 0.94 ($P < 0.0001$, $n = 602$), so we also used the more readily available MTCM as a measure of coldness.

Previous studies also suggest that cumulative winter temperature is important for the northward/upward distributions of warm-temperate and tropical tree species (Kira, 1948, 1991; Hattori & Nakanishi, 1985; Fang & Yoda, 1991), and for spring budburst of many northern tree species (Prentice *et al.*, 1992; Lechowicz, 2001). We used Kira's Coldness Index (CI) (Kira, 1948, 1991) to express the cumulative winter temperature:

$$\mathrm{CI} = -\sum (5 - T) \text{ (for months in which } T < 5\,^{\circ}\mathrm{C}) \qquad (3)$$

Annual mean temperature and mean temperature for the warmest month

Annual mean temperature (AMT) and mean temperature for the warmest month (MTWM) are often used to explain northward and upward distribution of northern tree species (Walter, 1979; Tuhkanen, 1980; Ohsawa, 1990, 1991).

Climatic continentality

Wolfe (1979), Tuhkanen (1980), Ohsawa (1990) and Fang *et al.* (1996) have addressed the effect of climatic continentality on distribution of some tree species. We used the annual range of monthly mean temperatures (ART) and Gorcynski's (1922) continentality index (K):

$$K = \left(1.7 \times \frac{R}{\sin L}\right) - 20.4 \qquad (4)$$

where R is the annual range of monthly mean temperature in °C and L is the latitude in degrees.

Moisture variables

To assess moisture regime, we considered annual precipitation (AP, mm), potential evapotranspiration (PET, mm), annual actual evapotranspiration (AAE, mm), moisture index (Im), and the Ellenberg quotient (EQ, °C mm^{-1}). With the exception of AP and EQ, all moisture parameters were estimated by the Thornthwaite (1948) method, which uses two climatic variables commonly recorded at climatic stations around the world: monthly mean temperature and monthly precipitation. The Thornthwaite index has proven a good correlate of vegetation and plant distribution at both regional and global scales (e.g. Mather & Yoshioka, 1968; Fang & Yoda, 1990; O'Brien, 1993; Frank & Inouye, 1994). For calculation of PET, AA and Im, see Fang (1989) and Fang & Yoda (1990).

The EQ is the ratio of the MTWM to annual precipitation and is frequently used to show the climatic limit of beech in Europe (Ellenberg, 1986; Jahn, 1991). According to Jahn (1991), values below 20 show a pure 'beech climate'; the competitive vigour of beech slowly decreases with an increase from 20 to 30; and in Europe beech disappears in regions with an EQ over 30:

$$\mathrm{EQ} = \frac{\text{Warmest month's mean temperature in } ^\circ\mathrm{C}}{\text{Annual precipitation (mm)}} \times 1000 \quad (5)$$

Principal components analysis

We used principal components analysis (PCA) (Wilks, 1995) to: (1) examine which climatic variables are most associated with the distribution of beech species, and (2) compare the climatic space at distributional limits for beech species in the three regions.

To assess the influence of the diverse climatic variables, we compared the eigenvalues of different PCA axes for each species and the axis loadings of climatic variables for the species that have a large sample size. In general, variables with bigger loadings are considered more important in placement along PCA axes when variables are standardized to allow for differences in units and magnitude (Wilks, 1995). We analysed 13 climatic variables (AMT, WI, CI, ABT, MTWM, MTCM, ART, K, AP, PET, AAE, Im and EQ) using PC-ORD software (McCune & Mefford, 1999). To minimize the differences associated with differing units and magnitude, all the variables were standardized:

$$x_i' = \frac{x_i - \bar{x}}{\sigma} \quad (6)$$

where x_i and x_i' are the original and standardized value of a climatic variable for the ith site, $\bar{x}$ is average of the climatic variable for all the sites, and σ is the standard deviation of the climatic variable. To compare climate spaces among the beech species, we used the eigenvalues of the first two components for each beech site. To illustrate the climate space for species with sufficiently large sample sizes, we calculated a 50% Gaussian bivariate confidence ellipse (ELL) using SYSGRAPH (SYSTAT Inc., 1996).

RESULTS

Climatic statistics at the distribution limits

Table 2 lists the average for thermal and moisture climatic parameters at lower/southern and upper/northern limits for all beech species; the details for the statistics (average, SD and range) of these variables are displayed in Table S1 in the Supplementary Material. The descriptions that follow are supported by both Table 2 and Table S1.

In East Asia, in spite of large fluctuations within and among species, average values of growing season warmth at the lower limits of beech distribution were between 74.8 °C (*F. crenata*) and 115.9 °C·month (*F. longipetiolata*) for Warmth Index (WI) and between 9.8 and 14.3 °C for annual biotemperature (ABT) (Table 2). Interestingly, most of the seven Asian beech species had lower limits associated with very similar average thermal parameters (AMT of 11.7–12.8 °C, WI of 91.8–100.6 °C·month, ABT of 11.7–12.8 °C, MTWM of 22.4–23.9 °C and MTCM of 0.1–2.1 °C) even though the species have widely disparate ranges (*F. japonica* in Japan, *F. multinervis* in Korea and others in China). Only two species, *F. crenata* from colder climates and *F. longipetiolata* in warmer regions, are exceptions. The thermal variables at the upper limit of East Asian beeches showed smaller fluctuation for most species with an AMT value of 7.3–9.2 °C, WI of 56.0–70.3 °C·month, and ABT of 8.1–9.5 °C (Table 2), but *F. crenata* is an exception. *Fagus crenata* has a WI value of 45.1 °C·month, approximately the climatic threshold (45 °C·month) of the cool-temperate zone defined by Kira (1991).

With regard to moisture regime, all beech species locations in East Asia fall within an average Im range of 61.4–189.9 for their lower limits and 70–241.3 for their upper limits, denoting a humid or perhumid climate according to the Thornthwaite (1948) system (Table 2).

In North America, *F. grandifolia* showed the widest climatic space, with average WI ranging from 50.7 °C·month (northern limit) to 173.4 °C·month (southern limit), and ABT from 7 to 19.5 °C. This spans two bioclimatic zones: cool-temperate (45–85 °C·month for WI and 6–12 °C for ABT) and warm-temperate zone (85–180 °C·month for WI and over 12 °C for ABT). Although *F. mexicana* is distributed much farther south than *F. grandifolia*, it has a smaller climatic range (117.1–127.3 °C·month in WI and 14.8–15.6 °C in ABT) and much smaller thermal variables at its lower limit than those of *F. grandifolia* (Table 2). *Fagus grandifolia* has more moist conditions at its northern limit with an average Im value of 91.1 compared with 40.3 at its southern limit. Mexican beech has a much larger Im average value (Im = 103) at its elevational and southern limit.

In Europe, the climatic spaces of beech species span almost the entire temperate zone. Growing season warmth in the range of beech ranged from 47.7 to 104.3 °C·month for WI and 7.2–13.5 °C for ABT for *F. sylvatica*, and 46.3–78.3 °C·month for WI and 7.1–10.4 °C for ABT for *F. orientalis* (Table 2). It is noteworthy that *F. sylvatica* has a rather narrow climatic range despite having the widest latitudinal range among the world beech species (spanning *c.* 23°) (Jalas & Suominen, 1972–91). In comparison with beech species in other regions, the moisture regime in the area of European beech species is relatively dry, with average precipitation of 905.9 mm, mean Im value of 38, and mean EQ of 29 °C mm^{-1} for the lower limit of *F. sylvatica*, and 1272.3 mm, 119.3 and

Table 2 Average for climatic variables at lower (southern) and upper (northern) limits of distribution for world beech (*Fagus*) species. AMT: annual mean temperature; WI: warmth index; CI: coldness index; ABT: annual biotemperature; MTWM: mean temperature for the warmest month; MTCM: mean temperature for the coldest month; ART: annual range of mean temperature; K: continentality index; AP: annual precipitation; PET: annual potential evapotranspiration; AAE: actual annual evapotranspiration; Im: moisture index; and EQ: Ellenberg quotient. – indicates no estimation

Species	Lower/upper limits	AMT (°C)	WI (°C·month)	CI (°C·month)	ABT (°C)	MTWM (°C)	MTCM (°C)	ART (°C)	*K*	AP (mm)	PET (mm)	AAE (mm)	Im	EQ (°C mm^{-1})
F. engleriana	Lower	12.4	99.0	−10.1	12.5	23.0	1.1	21.9	53.3	1230	805.4	803.3	61.4	21.7
	Upper	7.3	56.0	−28.1	8.1	17.8	−3.9	21.7	52.8	1366	795.6	793.5	70.0	19.4
F. hayatae	Lower	12.8	99.4	−6.4	12.8	22.4	2.1	20.3	50.7	1670	914.6	914.6	85.5	17.0
	Upper	9.2	67.0	−16.6	9.5	18.6	−0.7	19.3	46.3	2233	887.0	887.0	109.8	13.2
F. longipetiolata	Lower	14.3	115.9	−4.1	14.3	24.2	3.5	20.8	54.1	1421	845.4	831.8	77.5	19.0
	Upper	8.9	66.2	−19.8	9.2	18.9	−2.0	20.9	54.2	1738	621.2	619.2	132.6	14.7
F. lucida	Lower	12.7	100.6	−8.2	12.8	23.0	1.5	21.5	57.8	1419	874.0	862.7	70.0	19.5
	Upper	8.8	66.4	−21.3	9.2	19.1	−2.4	21.5	57.8	1757	674.4	671.7	168.5	13.8
F. crenata	Lower	9.5	74.8	−21.3	9.8	21.8	−2.1	23.9	48.5	2047	706.2	706.2	189.9	12.5
	Upper	5.0	45.1	−45.2	6.7	17.6	−6.8	24.4	48.8	2660	606.8	606.8	241.3	9.5
F. japonica	Lower	11.7	91.8	−12.0	11.7	23.9	0.3	23.7	48.8	1805	746.6	746.6	142.0	15.3
	Upper	9.0	70.3	−21.9	9.4	21.2	−2.2	23.4	48.6	2329	716.4	716.4	206.5	11.5
F. multinervis	Lower	11.5	90.2	−11.9	11.5	23.4	0.1	23.3	45.0	1485	702.6	702.6	111.3	16.1
	Upper	7.6	59.0	−27.5	8.3	19.5	−3.8	23.3	45.0	–	–	–	–	–
F. grandifolia	Lower	19.5	173.4	0	19.5	27.5	10.4	17.1	36.8	1426	1024.0	996.9	40.3	19.5
	Upper	4.2	50.7	−60.5	7.0	18.4	−11.4	29.8	49.5	1021	537.8	530.5	91.1	18.5
F. mexicana	Lower	15.6	127.3	0	15.6	19.6	10.5	9.1	22.2	1741	949.0	906.0	103.1	14.5
	Upper	14.8	117.1	0	14.8	18.7	9.7	9.1	22.2	–	–	–	–	–
F. sylvatica	Lower	13.5	104.3	−2.7	13.5	23.0	4.7	18.2	27.7	906	749.8	497.1	38.1	29.0
	Upper	6.6	47.7	−28.3	7.2	16.9	−2.7	19.6	25.8	1272	577.6	496.7	119.3	16.8
F. orientalis	Lower	10.2	78.3	−16.2	10.4	20.5	−1.5	22.0	38.1	745	717.2	486.5	15.4	32.7
	Upper	6.5	46.3	−28.5	7.1	16.1	−3.2	19.3	31.5	912	668.9	460.3	23.8	27.9

16.8 °C mm^{-1} for AP, Im and EQ for its upper/northern limit. For *F. orientalis*, AP, Im and EQ were, respectively, 744.8 mm and 32.7 °C mm^{-1} for its lower limits, and 911.8 mm, 23.8 and 27.9 °C mm^{-1} for its upper limit. It is apparent that climate at the lower limits of both beeches is rather dry, close to the EQ threshold of 30 °C mm^{-1} suggested by Jahn (1991). The EQ value of European beeches is much smaller than for beech species in other regions: ranging from 15 to 20 °C mm^{-1} for the lower/southern limit and from 11.5 to 19.4 °C mm^{-1} for the upper/northern limit for other regions, with the smallest value (12.5 and 9.5 °C mm^{-1} for its lower and upper limits, respectively) for *F. crenata* (Table 2).

Climatic factors controlling beech species distributions

We used the 13 climatic variables in a PCA to identify limiting factors for beech distribution. To assess regional variations in the relationships between geographic distribution and climatic space, we combined four Chinese beeches and two European beeches into the same group because they have similar climate ranges (cf. Table S1). The two Japanese beeches occur primarily in two different climatic regions (Japan Sea side and Pacific Ocean side), and thus we did not deal with them as a single group. The first two principal components account for 70% and 68% of overall variance for southern/lower and northern/upper limits of all species, respectively (Table 3). Accordingly we considered the loadings on these axes to be most important in delimiting beech distribution. For each beech region, the first two PCA axes explained more than 70% of the variance; for example, 80% and 87% for the southern and northern limits of Amerian beech, respectively, and 73–80% for *F. japonica* and the European beeches.

The first PCA axis represents a thermal gradient, and the second a moisture gradient in the overall distribution of world beech species (Table S2 in Supplementary Material). In general the thermal climate played a leading role and precipitation a secondary role in controlling the large-scale distribution of beech species. Among thermal variables, AMT, WI and ABT always showed larger loadings at both lower/southern and upper/northern limits, but CI and MTCM exhibited a large value for world beech species (Table S2). This indicates that growing season warmth is most important for beech distribution, and CI and MTCM are also closely coupled with the potential for their range expansion. The second PCA axis exhibited a different trend: for the lower/southern limit the loadings of AP, Im and EQ were 0.83 mm, 0.84 and −0.89 °C mm^{-1}, while for the upper limit/northern they were −0.83 mm, −0.92 and 0.89 °C mm^{-1}, respectively (Table S2). The negative loadings of AP and Im indicate that altitude limits for beech species are negatively correlated with moisture climate, implying that the altitude of the upper elevational limit decreases with an increase of precipitation.

In spite of these overall trends, some component loadings varied across regions, suggesting that limiting climatic factors shifted somewhat among beech in different regions. In general, both the lower/southern and upper/northern limits in all three beech regions depended on seasonal thermal regimes (loadings of growing season warmth (WI and ABT) on the first PCA axis were largest, and winter temperatures (CI and MTCM) also showed a large influence), but the two Japanese beech species were an exception. For the second PCA axis, the loadings of moisture climate variables were largest for the lower/southern limit, indicating the general influence of a moisture gradient. However, for the upper/northern limit, the climatic variables with the largest loading showed a large regional difference. The second axis for Chinese beech suggests a moisture regime gradient, while the loading of winter coldness (CI) was largest for European beeches, and MTCM had the largest loading for American beech.

For the lower limit of the two Japanese beeches, winter temperature (CI and MTCM) had a large loading on the first axis, summer temperature (MTWT) on the second. This suggests that moisture regime is not a limiting factor for the downward distribution of these two species due to abundant precipitation in Japan, which agrees with other studies (e.g. Maeda, 1991). For the upper limit of these two species, on the second axis, the loadings of PET and AAE for *F. crenata* and of MTWT and moisture indices (Im and EQ) for *F. japonica* were largest.

It is noteworthy that PET, an indicator of total solar energy, showed the largest loading (0.98) on the first axis for the northern limit of American beech; this is consistent with the correlation between PET and vegetation distribution and overall tree species richness in North America (Stephenson, 1990; Francis & Currie, 2003).

Table 3 Proportion (%) of cumulative variance on the first four principal components in a principal components analysis of the distribution limits for world beech (*Fagus*) species

Species	Lower (southern) limit				Upper (northern) limit			
	PCA 1	PCA 2	PCA 3	PCA 4	PCA 1	PCA 2	PCA 3	PCA 4
Chinese beeches	46.34	69.89	83.71	96.69	41.19	73.56	89.04	96.12
F. crenata	37.60	71.86	90.25	97.27	50.48	74.57	88.08	96.49
F. japonica	47.98	72.96	88.70	98.34	49.27	80.20	90.85	97.71
F. grandifolia	59.25	79.76	91.69	98.77	54.46	86.65	96.99	99.12
European beeches	43.65	76.86	87.73	94.73	40.45	71.79	88.01	97.26
All beech species	50.71	69.89	84.98	94.28	44.89	67.57	83.72	93.79

DISCUSSION

Zonal distribution of world beech species

Although most beech species are considered typical trees of the temperate zone, they in fact showed different climatic ranges in the three regions where they occur (Table 2; Table S1). In East

Asia, most species, excluding *F. crenata*, concentrated within a climatic range of 90–116 °C·month in WI and 11.7–14.3 °C in ABT in their lower limits, and 56–70 °C·month in WI and *c.* 8.1–9.5 °C in ABT in their upper limits. This is to say, the boundaries for most Asian beech species are located in warmer places than the cool-temperate zone defined by Kira (1945, 1991) and Holdridge (1947); the climatic parameters at the southern/lower edge of the cool-temperate zone was set at 85 °C·month in WI by Kira, and 12 °C in ABT by Holdridge, and 45 °C·month in WI and 8 °C in ABT at the northern or upper limit. This suggests that beech species occupy an ecotone between cool-temperate deciduous broadleaf forest (a WI range of 45–85 °C·month) and warm-temperate evergreen broadleaf forest (WI > 85 °C·month) (Kira, 1991). However, the WI value at the upper limit of *F. crenata* in Japan (45.1 °C·month) coincided well with Kira's criterion of 45 °C month; this supports the idea that *F. crenata* is an indicator of the Japanese temperate zone (e.g. Miyawaki, 1980–89).

American beech occupies a large climatic space, with a range of 50.7–173.4 °C·month in WI and 7.0–19.5 °C in ABT (Table S1). This spans two bioclimatic zones: the cool-temperate zone [45–85 °C·month for WI (Kira, 1945, 1991) and 6–12 °C for ABT (Holdridge, 1947)], and the warm-temperate zone (85–180 °C month for WI and > 12 °C for ABT).

In Europe, climatic parameters at the northern/upper limit of beech distribution indicated good agreement with criteria defining the cool-temperate zone: a WI value of 47.7–104.3 °C·month, and an ABT value of 7.2–13.5 °C for *F. sylvatica*, and 46.3–78.3 °C·month and 7.1–10.4 °C for *F. orientalis* (Table S1). On the other hand, an average Im greater than 15.4 for all beech sites suggest a perhumid or humid climate, defined by Thornthwaite (1948) as Im values of 20–100 and > 100, respectively.

Climates and present distribution of East Asian beeches

As shown in Table 3, the first two PCA components account for more than 70% of the variance in parameters associated with the distribution of Asian beech species. No one variable alone can explain beech distributional patterns; growing season warmth (WI and ABT), winter low temperature (CI) and annual mean temperature (AMT) all showed almost equal loadings (Table S2). Thermal regime is clearly paramount in the relationships between beech distribution and climatic factors in East Asia, but the strong correlation among different climatic elements (see Table S3 in Supplementary Material) precludes identification of a single, dominant aspect of thermal regime that affects the distribution of East Asian beech species.

There are a few viewpoints on the relationships between the present distribution of Chinese beech species and limiting factors. Hong & An (1993) pointed out that the climatic factors affecting beech distribution varied from place by place; for example, in northern regions, coldness and short growing season were major limiting factors, whereas water deficit was more important for southward migration. Cao *et al.* (1995) demonstrated the importance of the moisture deficit in the northern range of beech species and the importance of high temperature and insufficient water supply in the south. Focusing on the relationship between *Fagus*- and *Tsuga*-dominated forests, Fang *et al.* (1996) suggested possible effects of the annual temperature range (ART) and high winter temperature on beech distribution. They found that the beech-dominated forests did not appear in places where hemlock dominates, and that their boundary was consistent with an ART isotherm of 23 °C; beech-dominated forests lay north of this isotherm and hemlock-dominated forests lay south. This implies the importance of high winter temperatures in determining the distribution of Chinese beech because the ART is closely correlated with winter temperature in southern mountain areas in China.

Two effects of high winter temperatures that can influence the distribution of temperate tree species may be playing a role in beech distribution in China. First, temperate trees require a sufficient period of winter cold (a period of chilling) before warming will induce budburst in spring (Cannell & Smith, 1986; Lechowicz, 2001). Second, higher winter temperatures can reduce the competitive ability of deciduous trees with more warmth tolerant evergreen broadleaf trees (Woodward, 1987). These effects of winter temperature may explain an unresolved question in Asian biogeography: why beech species do not spread westward into the Himalayas and south-eastern Tibet where growing season warmth and precipitation appear satisfactory. Although Asian beech and hemlock have similar heat requirements during the growing season (Liu & Qiu, 1980; Hou, 1983; Fang *et al.*, 1996), hemlock is favoured by a warm-winter climate (Sakai, 1975). While Chinese beeches co-exist with evergreen broad-leaved tree species in genera such as *Lithocarpus*, *Cyclobalanopsis* and *Castanopsis* (Wu, 1980; Hou, 1983; Fang, 1999), there may be a point where a warm-winter climate tips the competitive balance to evergreen tree species. Observations from Woodward (1987), Cao (1995), Williams-Linera *et al.* (2000) and Miyazawa & Kikuzawa (2005) support this hypothesis that warm winters can favour evergreens and limit range expansion in beech species.

In Japan, growing season warmth rather than winter temperatures is generally seen as the control on the distributions of beech species (Kira, 1945; Miyawaki, 1980–89; and many others), and Japanese cool-temperate vegetation is sometimes termed the beech forest zone (e.g. Miyawaki, 1980–89). Precipitation is also used to explain differences in community composition and structure of Japanese beech forests (Kure & Yoda, 1984; Hattori & Nakanishi, 1985; Fang & Yoda, 1990; Matsui *et al.*, 2004). In particular, distribution of *F. crenata* along the Pacific Ocean side and the Japan Sea side of Japan usually has been explained by accumulated snowfall (Yamazaki, 1983; Maeda, 1991; Matsui *et al.*, 2004), with pure beech forests found only on the Japan Sea side where snowfall is extremely abundant. Given the situation in China, however, we should not discount out of hand the possible influence of low winter temperatures in the development of pure beech forests on the Japan Sea side. Winter temperature there is much lower than on the Pacific Ocean side; this could strengthen the competitive ability of beech against other tree species (*Tsuga* and *Quercus*) and understorey bamboo (*Sasa*), and enable its dominance.

A question remains with regard to Japanese beech distribution: why is there no beech on Yakushima Island (30°27′ N, 130°30′ E, 1935 m a.s.l.) in southern Japan where the climate is suitable for beech growth and there are many species that co-occur with beech in Kyushu, Shikoku and Honshu (Miyawaki, 1980–89). This may be related to some topographic barriers or climatic limitations at the time during glaciation when Yakushima was part of the major Japanese islands, or simply to dispersal limitation since sea levels rose and isolated the island. Another possibility worth investigating is that a combination of warm winters and a photoperiodic influence on the timing of budburst in beech (Falusi & Calamassi, 1996) lead to poor synchrony between spring budburst and the early part of the growing season that puts beech at a competitive disadvantage.

Climates and the present distribution of American beech

The PCA showed that MTCM and PET have almost equivalent loadings to growing-season warmth (WI and ABT) for the southern limit of *F. grandifolia*, whereas heat (WI and ABT) and energy (PET) are most important for the northern limit (Table S2). Although Huntley *et al.* (1989) demonstrated the role of January and July mean temperatures in the present distribution and abundance of beech species in North America and Europe using pollen percentages of surface samples, they used only monthly mean temperatures as thermal parameters. Our findings based on survey of many more climatic parameters are generally consistent with their conclusions.

Although our study suggested that the growing season warmth was associated with the northerly distribution limit, some physiological observations emphasize the adverse effects of excessively low winter temperatures for many temperate North American trees (Sakai & Weiser, 1973; Hicks & Chabot, 1985; Denton & Barnes, 1987; Maycock, 1994). Many studies stress the influence of low winter temperature, but perhaps only because it is easier to do experimental manipulations of chilling effects than warmth during the growing season. We can, however, consider biogeographic evidence supporting the importance of growing season temperature. The lack of American beech in Newfoundland, Canada, where winter temperatures are much higher than at the northern edge of beech distribution (Table 4) suggests growing season warmth is a greater limitation than winter cold. Climatic statistics of thermal variables at 18 stations located between 47° and 48° N in Newfoundland where the latitudes were coincident with the northern limit in east Quebec show far higher winter temperatures (CI and MTCM) in Newfoundland than at the beech northern limit (in the former, −34.8 °C·month for CI and −4.2 °C for MTCM, and the latter −60.5 °C·month and −11.4 °C). In contrast, the growing season temperatures were much lower in Newfoundland than at the beech northern limit; for the former, WI, ABT and MTWM were 36.7 °C, 6.1 °C and 15.6 °C·month, and for the latter those are 50.7 °C, 7.0 °C and 18.4 °C·month, respectively (Table 4). This suggests that insufficient warmth during the growing season may be a factor limiting the expansion of beech into Newfoundland. This hypothesis is supported by eco-physiological studies of flowering and seed production showing that a certain minimum degree of heat is required for floral initiation, and flower and seed production in many temperate tree species (Matyes, 1969; Owens & Blake, 1985). The comparative study of masting behaviours of beech species also shows the importance of summer heat in controlling beech seed production (Piovesan & Adams, 2001).

Table 4 Thermal variables for Newfoundland, Canada, based on 18 climatic stations (Atmospheric Environmental Service, Environment Canada, 1982). For comparison, the estimated mean value of climate variables at the northern limit of *Fagus grandifolia* in North America ('Mean at northern limit' column) is also tabulated. See Table 2 for abbreviations for climatic variables

Variable	Mean	SD	Minimum	Maximum	Mean at northern limit
AMT (°C)	5.2	0.5	4.1	5.8	4.2
WI (°C month)	36.7	3.0	30.2	41.6	50.7
CI (°C month)	−34.8	4.2	−42.2	−26.3	−60.5
ABT (°C)	6.1	0.3	5.3	6.4	7.0
MTWM (°C)	15.6	0.5	14.7	16.4	18.4
MTCM (°C)	−4.2	1.1	−6.1	−1.9	−11.4
ART (°C)	19.8	1.1	17.2	22.5	29.8
K	25.3	2.6	19.5	31.4	49.5

Although temperature can account for the distribution limits of American beech, Fig. 1 suggests that continentality (*K*) also is important for limiting the southern and northern distribution limits. The fact that WI and MTCM at the southern limit decrease markedly with increasing *K* value shows that beech requires more heat and higher winter temperature in an oceanic climate than in a continental one. However, growing season warmth tends to increase as *K* increases at the northern limit (Fig. 1a), suggesting higher summer temperatures in the continental than in the oceanic climate. The relationship between winter temperature and *K* values at the northern limit shows the same pattern as at the southern limit (Fig. 1b). Similar results were found for the distributions of some tree species and vegetation zones in East Asia (Ohsawa, 1990; Fang & Yoda, 1991; Fang *et al.*, 1996).

Climates and present distribution of beech species in Europe

The relationships between beech distribution and environments in Europe have been discussed from the viewpoint of soil, topography and climate (Ellenberg, 1986; Jahn, 1991).

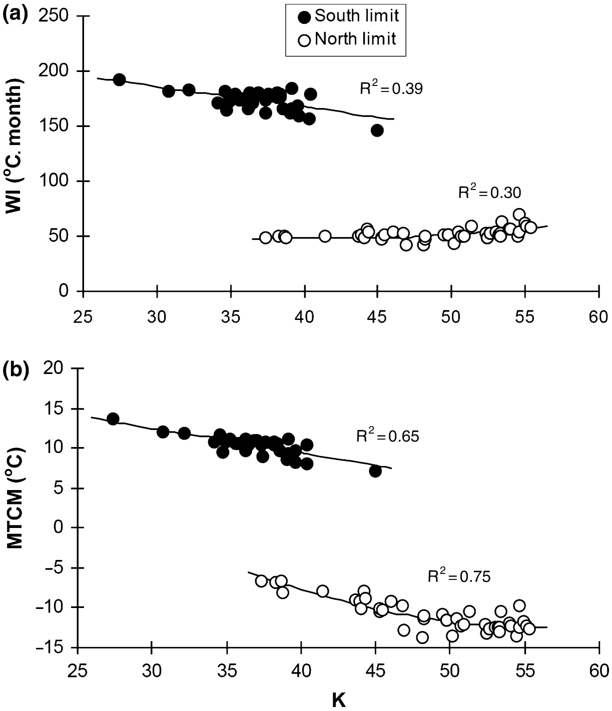

Figure 1 Relationships between (a) warmth index (WI) and (b) mean temperature for the coldest month (MTCM) and continentality index (K) at the southern (filled circles) and northern (open circles) limits of *Fagus grandifolia* in North America. The relationships for the northern limits are fit by a nonlinear regression.

These earlier studies supported the importance of growing season temperatures, but did not focus in any detail on the array of thermal parameters that might be involved. Both thermal climate and continentality (K) contributed to European beech distributions, much as in North America. Figure 2 expresses the relationships between the thermal variables (WI and MTCM) and K value at the northern limit of *F. sylvatica*. With an increase of the K values, WI increased (Fig. 2a), and MTCM decreased (Fig. 2b).

Although beech has been present in south-eastern England since at least 3000 yr BP (Birks, 1989), its present distribution in the British Isles does not appear to be in equilibrium with present climate. Climatic analysis shows that beech still has not reached its potential northern range limit (Table 5). Compared with averages of climatic variables at the upper/northern limits on the European mainland (AMT of 6.6 °C, WI of 47.7 °C·month, and ABT of 7.2 °C), those at the present northern limit in Britain were much larger (more southerly), by *c.* 3.4 °C for AMT, 15 °C·month for WI and 2.8 °C for ABT (Table 5). Assuming a decrease in mean temperatures of 0.5 °C per degree latitude in higher latitudes (Fang, 1996), beech should arrive at its range limit *c.* 6° to the north of its present range boundary. Spreading north at a rate of 100–200 m yr^{-1} (Birks, 1989), beech should reach its potential natural distribution in Britain 3500–7000 years in the future if one assumes an unchanging climate scenario. This implies that most of Britain eventually should be covered by beech.

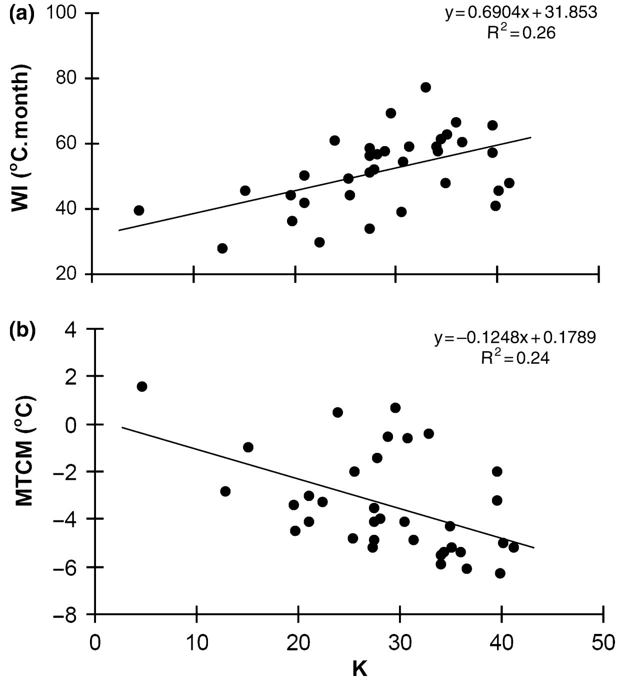

Figure 2 Relationships between (a) warmth index (WI) and (b) mean temperature for the coldest month (MTCM) and continentality index (K) at the northern limit of *Fagus sylvatica* in Europe.

Table 5 Climatic parameters at the northern limit for *Fagus sylvatica* in England based on 11 climatic stations. For comparison, the estimated mean value of climate variables at the northern limit of *F. sylvatica* on the European mainland ('Mean at northern limit' column) is also tabulated. See Table 2 for abbreviations for climatic variables

Variables	Mean	SD	Minimum	Maximum	Mean at northern limit
AMT (°C)	10.0	0.4	9.1	10.7	6.6
WI (°C month)	62.6	3.4	53.8	69.2	47.7
CI (°C month)	−2.3	1.2	−4.2	0.0	−28.3
ABT (°C)	10.0	0.4	9.1	10.7	7.2
MTWM (°C)	16.8	0.4	15.8	17.6	16.9
MTCM (°C)	3.9	0.6	3.1	5.5	−2.7
ART (°C)	12.9	0.7	10.6	13.9	19.6
K	7.7	1.6	2.8	9.8	25.8
AP (mm)	707.2	148.4	540.0	1068.0	1272.3
PET (mm)	646.8	8.6	625.0	661.0	577.6
AAE (mm)	560.6	44.3	497.0	652.0	496.7
Im	14.7	20.4	−8.8	64.2	119.3
EQ (°C mm^{-1})	24.7	4.8	15.4	32.4	16.8

Comparison of climatic space for world beech species

The present distributions of most tree species are strongly related to climate (Woodward, 1987; Huntley *et al.*, 1989; Francis & Currie, 2003). Simple isotherm methods have long been used to assess distributional limits in relation to climate (Hutchinson, 1918; Koppen, 1936; Kira, 1945, 1991; Thornthwaite, 1948;

Bryson, 1966; Tuhkanen, 1980; Grace, 1987; Arris & Eagleson, 1989; and many others), but the nuances of climatic controls on range limits are not assessed fully by isotherms. In this study, we used a more comprehensive PCA approach to detect the single and joint importance of diverse climatic parameters in explaining the distribution of beech species. We are thus able to compare the climatic space of beech distribution among species in the three separate geographic regions where beech are major components of forest ecosystems.

Because the first two PCA axes were most important in explaining the distribution of beech species around the world, sample scores of these axes were used to compile scatter diagrams comparing the climatic spaces (climatic niches) of the respective beech species (Figs 3 and 4). Because of a large sample size and rather scattered score values, 50% Gaussian bivariate confidence ellipses (ELL) were drawn for the species with a large sample size to more easily compare climatic conditions among beech species. Also, because four Chinese beeches and two European beeches have similar moisture and warmth requirements (see Table 2; Table S1), sample data were combined.

Figures 3 and 4 show climatic gradients associated with the lower (southern) limit and the upper (northern) limit of beech. At its lower or southern limits (Fig. 3), *F. grandifolia* extends to warmer regions, while *F. crenata* requires less warmth, but both Japanese beeches occur in more moist climates than *F. grandifolia*. Chinese and European beeches occupy similar temperature ranges, but the latter is in a drier climate. At the upper or northern edges (Fig. 4), both *F. grandifolia* and *F. crenata* occupy colder areas, while Chinese beeches have similar warmth demand to *F. japonica*. Along the moisture axis, *F. crenata* occurs in the most humid conditions, and European beech in the driest habitats. Although the actual influence of climatic factors on species distributions may be nonlinear and only partly reflected in the present analyses (Austin, 2002), it is clear that there is a degree of climatic niche differentiation among the extant beech species.

CONCLUSIONS

Focusing on distribution limits of the world beech species, we compare their climatic spaces in three different regions globally, and explore the climatic correlates of these distribution patterns. The results suggest that thermal climate is most important overall in determining the distribution of beech

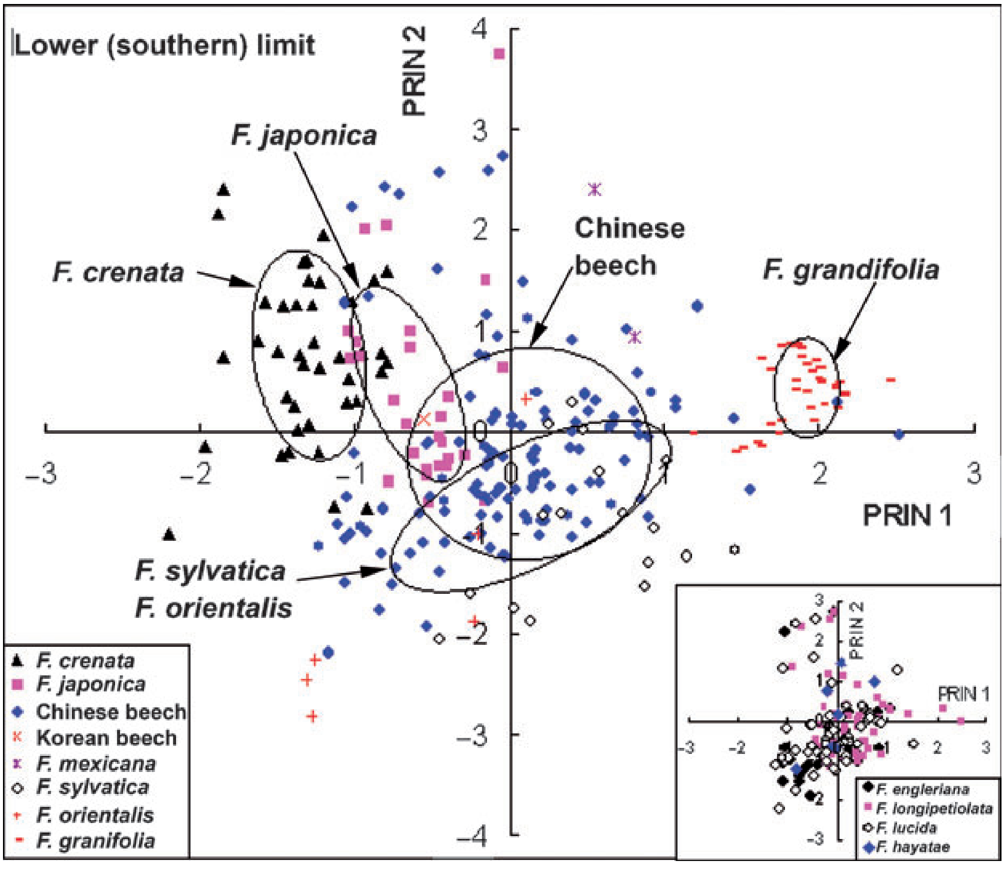

Figure 3 Climatic spaces for world beech (*Fagus*) species at their lower/southern distribution limit. The first two principal components are plotted. The first axis indicates a gradient in thermal climate and the second a moisture gradient in the overall distribution of world beech species. The 50% ELL is drawn for major beech species to show their primary ranges. Inset graph shows climatic scores of four Chinese beech species that have similar climatic ranges.

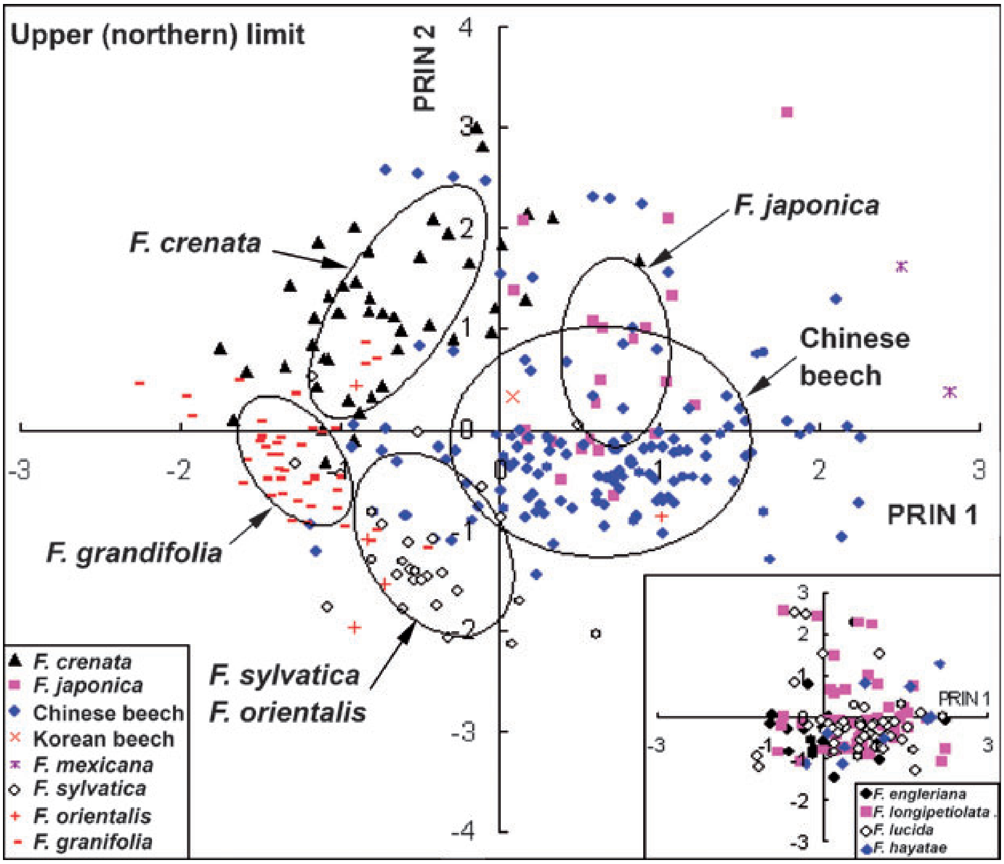

Figure 4 Climatic spaces for world beech (*Fagus*) species at their upper/northern distribution limit. The second axis, indicated by precipitation and moisture index for most beech species, shows a negative correlation with beech distribution (Table S2 in Supplementary Material); we have inverted the scale of this axis to more easily compare with Fig. 3. The occurrence of *F. grandifolia* to the lower-left does not indicate its northern limit is set by dry conditions; the second axis indicates low winter temperature, but not moisture regime, with the loadings of −0.98 and −0.91 for MTCM (mean temperature for the coldest month) and CI (coldness index) (Table S2), respectively. For additional explanations see Fig. 3.

species, and that moisture effects are secondary. The degree and duration of low winter temperature (MTCM and CI), and annually available solar energy (PET) sometimes also played a role. At the lower or southern limits, *F. grandifolia* occurred in much warmer regions, and *F. crenata* in colder regions; Chinese and European beeches have similar, intermediate heat requirements. Along moisture gradients, Japanese beeches appeared in more moist conditions and Chinese and European beeches in drier situations. At the upper or northern limits, *F. crenata* and American beeches had similar, relatively low warmth demands, while *F. japonica* and Chinese beeches were found only in warmer regions. Along a moisture gradient, the Japanese beech species again occupied the most moist regions, with European beech in contrast occupying the most dry (Fig. 4).

Growing season temperature was most important in explaining overall distribution of Chinese beeches, but their northern limits were mainly set by low precipitation. The climatic factor controlling their westward expansion (south-east Tibet and Himalaya) may be higher winter temperatures that influence their budburst in spring and weaken their competitive ability with evergreen hemlock and broad-leaved evergreen trees. Although the distribution limits of beech species in Japan were controlled by summer temperature, their dominance may depend on regional climatic factors such as snowfall and winter low temperature. Winter low temperature may enhance the competitive ability of *F. crenata* with other co-existing species, allowing it to form pure beech forests in western Japan.

High summer temperature was considered to be the limiting factor for southward extension of American beech, while adequate growing season warmth was critical for its northward distribution. Continentality (*K*) played an important part in delimiting its range expansion, but lack of growing season warmth was the most important climatic factor precluding its migration to the Atlantic Islands (such as Newfoundland, Canada). Summer temperature is a limiting factor for the distribution of beeches in Europe, but continentality was also associated with limits to their north-western distribution. The northerly distribution of beech in Britain has apparently not reached its potential limit due to lack of time since deglaciation.

Although the present-day distribution patterns of beech species showed good correspondence to contemporary climate, isolated exceptions exist. For example, despite favourable climatic conditions there are no records of beech ever having grown on Yakushima Island in Japan. Similarly, *F. mexicana*, isolated in a small mountainous area of north-eastern Mexico, shows no clear association with contemporary climate. Therefore, a historical view on beech distribution is an essential complement to the climatic analyses emphasized in this paper.

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SUPPLEMENTARY MATERIAL

The following supplementary material is available online from http://www.Blackwell-Synergy.com

Appendix S1 Location and elevation of distribution limits of beech (*Fagus* L.) species in the present study.

Table S1 Statistics for climatic variables at lower (southern) and upper (northern) limits of distribution for world beech species.

Table S2 Loadings of climatic variables derived from Principal Components Analysis for the first three principal components associated with the distribution limits of world beech species.

Table S3 Coefficient of correlation between annual mean temperature (AMT) and other thermal variables across the global range of beech species.

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